MODEL FAIRNESS PASSPORT · AUDIT BEYOND ACCURACY

Brain-age model SFCN · IXI

An independent audit of accuracy, group fairness, acquisition robustness, explanations and uncertainty.

Passport no. **BAF-XAI-2026-001 v2**
Issued **2026-10-26**
Checkpoint **sha256 13ddd261…**
Valid for **IXI-like T1 MRI**

ARCHITECTURE	PRETRAINING	FINE-TUNED ON	EVALUATED ON
SFCN, 3D CNN 72 age bins (18–90)	**UK Biobank** Peng et al. 2021	**IXI, 339 adults** 3 London sites	**73 test + 412 CV** test used once

PASS Accuracy · **PASS** Fairness by sex · **FAIL** Fairness by site · **FAIL** Scanner invariance · **PASS** Faithful explanations · **FAIL** Honest uncertainty

PERFORMANCE & FAIRNESS

CHECK	RESULT	95% CI
● Test MAE	3.71 yrs · r 0.966	3.10 – 4.38
● Gap, male vs female (CV, equivalent)	+0.61 yrs	−0.50 – +1.72
● Gap, HH vs Guys (CV, concern)	−1.51 yrs	−2.64 – −0.38
● Gap, IOP vs Guys (test)	**−5.06 yrs**	−8.66 – −1.34
● Gap, IOP vs Guys (CV, 52 IOP)	**−5.24 yrs**	−7.56 – −2.92
● Extra error, IOP / HH (CV, concern)	+4.06 / +1.61 yrs	not on test
● IOP unseen (extra error)	**+3.30 yrs**	+1.62 – +4.93
● HH unseen (extra error)	+0.25 yrs	−0.88 – +1.36

ROBUSTNESS, EXPLANATIONS, UNCERTAINTY

CHECK	RESULT
● Site read from features	**91%** (chance 33%)
● Scan look alone shifts age	**3–4 yrs**, 100% same direction
● IOP offset explained by age, intensity, blur, head size, cohort, ComBat or age-conditional erasure	no, all persist
● IOP offset before any fine-tuning	−3.3 yrs, already present
● Sanity check (randomised layers)	ρ 0.13, maps destroyed
● Deletion test, top vs random 5% (2 baselines)	4.83 vs 1.18; 6.49 vs 1.99
● Own 90% range coverage, IOP	**77%** (Guys 94%)
● Per-site conformal 90% coverage	91–92% every site

USAGE GUIDANCE

SUITABLE FOR

- Research on brain ageing within a single acquisition site
- Comparing groups scanned on the same scanner and protocol
- Teaching and methods development

NOT SUITABLE FOR

- Any clinical or individual decision
- Comparing brain-age gaps *across* scanners without site-aware validation
- Uncertainty intervals that are not calibrated per site

KNOWN LIMITATIONS

- IOP has 61 subjects; the offset replicates in validation, test, 5-fold CV and three training seeds (−5.2 to −6.2).
- In IXI, site is confounded with who each site recruited: the cause of the IOP offset (untested scanner properties or cohort) cannot be separated.
- Healthy volunteers aged 20–86 only. Attribution resolution 8 mm. Uncertainty is the model's own spread, not a calibrated interval.

Verdicts follow criteria fixed in advance (**release/passport_criteria.json**) and are generated by **scripts/29**; evidence from **scripts/03b–28** and is logged with dates in **PROJECT_GUIDE.md**. Statistics: age-adjusted OLS with HC3 errors, 5,000-sample bootstrap CIs, FDR where several regions are tested. Wording: acquisition-site effects, not "hospital bias"; no clinical claims.

AUDITED beyond accuracy